FACULTY ELECTRICAL ENGINEERING, MATHEMATICS
AND COMPUTER SCIENCE

Department of Microelectronics

[ET4371] Mixed-mode Wireless transceivers

Assignment 2
DPLL Simulation

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Table of Contents

1 Part I [S-domain Simulation]

1.1 Answer on questions B, F, and bandwidth in C

The ADPLL is second order, since I have two integrators; the VCO and the loop filter. The damping factor and the natural frequency are:

$$\zeta = \frac{1}{2}\frac{a}{\sqrt{\rho}} \tag{1.1}$$

$$\omega_n = \sqrt{\rho} * F_{REF} \tag{1.2}$$

I chose a damping factor of $1/\sqrt{2}$. This is a common decision since it is balanced between stability and adequate response.

[note: I also swept ζ in matlab and found out that ζ=1 gave me the best performance, with the combination of an appropriate bandwidth, (around 7fs less jitter). Since the difference was small and I had already completed the theoritical/numerical analysis with this value, $\zeta = 1/\sqrt{2}$, I decided to keep the results (the deviation is small)]

By substituting this value is equation 1.3 we can find the relationship between the bandwidth and the natural frequency:

$$\omega_{-3dB}^2 = \omega_n^2\left((2\zeta^2+1)+\sqrt{(2\zeta^2+1)^2+1}\right) \tag{1.3}$$

$$\omega_{-3dB} = 2.06\omega_n \tag{1.4}$$

The decision of the bandwidth is based on the lowest possible jitter. The lowest jitter is achieved when the DCO contribution to the total phase noise is equal to that of the other loop components. From the description of the exercise we have 3 contributors of noise; the reference, the TDC and the DCO.

The reference noise is flat, and it follows a low-pass characteristic due to the transfer function from the reference to the output. The noise of the TDC, that its origin is its quantization noise, is also flat. Moreover, it follows a low-pass characteristic due to the transfer function from the TDC to the output. The slope of the phase noise for both of the noise contributors would be -20dB at the out of band, since there are two poles and one zero.

Lastly, there are three type of noise for the DCO. The flicker noise which have a -30dB characteristic, the thermal noise which have a -20dB characteristic and the quantization noise which also has a characteristic of -20dB. However, since we do not take into account the flicker noise in this exercise the slope in all the frequency spectrum of the DCO noise is -20dB. The transfer function of the DCO follows a high-pass characteristic with a 40dB slope meaning that inband it has a slope of +20dB and out of band a slope of -20dB.

The bandwidth should be big enough in order to suppress the noise of the DCO but not too big allowing the inband noise of the TDC and the reference to increase.

According to what it has been analysed until now, the DCO should be dominant at the out of band region since the loop suppress its contribution in band, while the TDC and the reference should be dominant in band due to their low-pass characteristic. According to the theory, the optimal bandwidth is this in which the DCO contribution to the total phase noise is equal to that of the other loop components. That means that the total power of the phase noise from the TDC and the reference should be equal with that of the DCO.

At this point the power spectral density of each contributor is analysed (question F). For the flat region of the reference at the output of the ADPLL:

$$L_{REF,OUT} = -155dBc/Hertz + 20log(24) = -127.4dBc/Hz$$
$$S_{REF,OUT} = 2 * 10^{-127.4/10} = 3.64 * 10^{-13} rad^2/Hz$$

The expression of the TDC power spectral density is derived from the slides:

$$S_{TDC} = \frac{(2\pi)^2}{12} * \frac{1}{F_{REF}} * \left(\frac{\Delta t_{res}}{T_{REF}}\right)^2 \tag{1.5}$$

By substituting:

$$S_{TDC,in} = 1.32 * 10^{-15} rad^2/Hz$$
$$L_{TDC,out} = 10log(0.5 * 1.32 * 10^{-15}) + 20log(24) = -124.2dBc/Hz$$
$$S_{TDC,out} = 2 * 10^{-124.2/10} = 7.6 * 10^{-13} rad^2/Hz$$

I add the power spectral densities of these two contributors in band and calculate what is the phase noise from 0 to 100MHz offset:

$$S_{total} = 7.6 * 10^{-13} + 3.64 * 10^{-13} = 11.24 * 10^{-13} rad^2/Hz$$
$$\sigma_\varphi^2 = \int_0^{f_{BW}} 11.24 * 10^{-13}\, df + \int_{f_{BW}}^{10^8} 11.24 * 10^{-13} \left(\frac{f_{BW}}{f}\right)^2 dx \Rightarrow$$

$$\sigma_\varphi^2 = 11.24 * 10^{-13} \left(2f_{BW} - \frac{f_{BW}^2}{10^8}\right) \tag{1.6}$$

The expressions that describe the characteristic of the DCO power spectral density are two. One inband (slope +20dB), one out of band (slope -20dB). The S_{peak} is the power spectral density at the bandwidth.

$$S_{in} = S_{peak} * \left(\frac{f}{f_{BW}}\right)^2$$

$$S_{out} = S_{peak} * \left(\frac{f_{BW}}{f}\right)^2$$

$$\sigma_{\varphi}^2 = \int_0^{f_{BW}} S_{peak} * \left(\frac{f}{f_{BW}}\right)^2 df + \int_{f_{BW}}^{10^8} S_{peak} * \left(\frac{f_{BW}}{f}\right)^2 dx \quad (1.7)$$

The S_{peak} can be found, by knowing what is the quantization noise and the thermal noise of the DCO at 1MHz. By integrating the frequency quantization noise I find the power spectral density of the DCO quantization noise. Moreover, I have to state that I did not use the sinc function in order to simplify my calculations. It is taken into account though in the matlab script and as a result I expect deviations from the ideal case where the DCO jitter contribution is equal to that of the other contributors.

$$S = |\frac{2\pi}{s}|^2 \frac{K_{DCO}^2}{12 f_{ref}} = \frac{0.5208}{f^2} rad^2/Hz$$

For 1MHz the DCO quantization noise is:

$$S = 5.21 * 10^{-13} rad^2/Hz$$

I add the power spectral densities coming for the quantization and thermal noise of the DCO at 1MHz and I find the peak value at the bandwidth. The peak value should be at the bandwidth because they both follow a 20dB characteristic in band and -20dB characteristic out of band

$$S_{total} = 5.21 * 10^{-13} + 2 * 10^{-120/10} = 25.21 * 10^{-13} rad^2/Hz$$

I calculate the S_{peak} in terms of the f_{BW}:

$$S_{peak} = 25.21 * 10^{-13} * \left(\frac{10^6}{f_{BW}}\right)^2$$

By substituting the S_{peak} value to the phase noise integral 1.7 and by equating the phase noise contribution of TDC and reference on one side and DCO on the other the bandwidth can be found as:

$$\boxed{f_{BW} = 1.23MHz}$$

Now I can calculate the loop filter coefficients from 1.1 and 1.2, since I know the natural frequency from 1.4 and I have selected the ζ.

$$\omega_n = \frac{1}{2.06} 2\pi * 1.23MHz = 3.75 * 10^6 radian/sec$$

$\rho = 14.06 * 10^{-4}$	$\alpha = 5.3 * 10^{-2}$

To find the integrated jitter, at first we find the phase standard deviation which is the square root of the integral of the power spectral density:

$$\sigma_{\varphi} = \sqrt{\int S_{total}(f)\, df}$$

To convert this into the standard deviation on time noise, known as jitter, we have to divide by the phase of clock that the PLL generates which is that if the DCO.

$$\sigma_\tau = \frac{\sqrt{\int S_{total}(f)\, df}}{2\pi F_{DCO}}$$

The calculation was fulfilled by the matlab code and the result is:

$$\boxed{\sigma_\tau = 174.3 fs}$$

1.1.1 Optimization

After executing the matlab code I noticed that the phase noise of the DCO was dominant compared to the phase noise of the reference and the TDC. That was expected, due to the approximation of the quantization noise of the DCO (I did not consider the sinc function).

For this reason, I swept the bandwidth in larger values to attenuate the dominance of the DCO. For a bandwidth of 1.78 MHz I observed the lowest possible jitter of:

$$\boxed{\sigma_\tau = 168.47 fs}$$

In this bandwidth, the contribution of the DCO and that of the TDC were equal as the theory suggest.

```
--- Integrated RMS Jitter Contributions ---  1.78 MHz
Reference:          67.91 fs
TDC Quantization:   97.96 fs
Combined Reference and TDC: 119.20 fs
DCO Thermal:        106.32 fs
DCO Quantization:   53.57 fs
Combined DCO and DCO Quantization: 119.06 fs
-------------------------------------------
Total RMS Jitter:   168.47 fs
Proportional Gain:   76.78 10^-3
Integral Gain:  2.95 10^-3
```

Figure 1.1: Jitter results.

The theoretical method used was validated, and also the approximation made during its implementation did not cause large deviation from the ideal bandwidth (for a common choice of $\zeta = 1/\sqrt{2}$):

$$\boxed{f_{-3dB} = 1.78 MHz}$$

The resulted phase noise of each individual contributor and the total jitter is presented below.

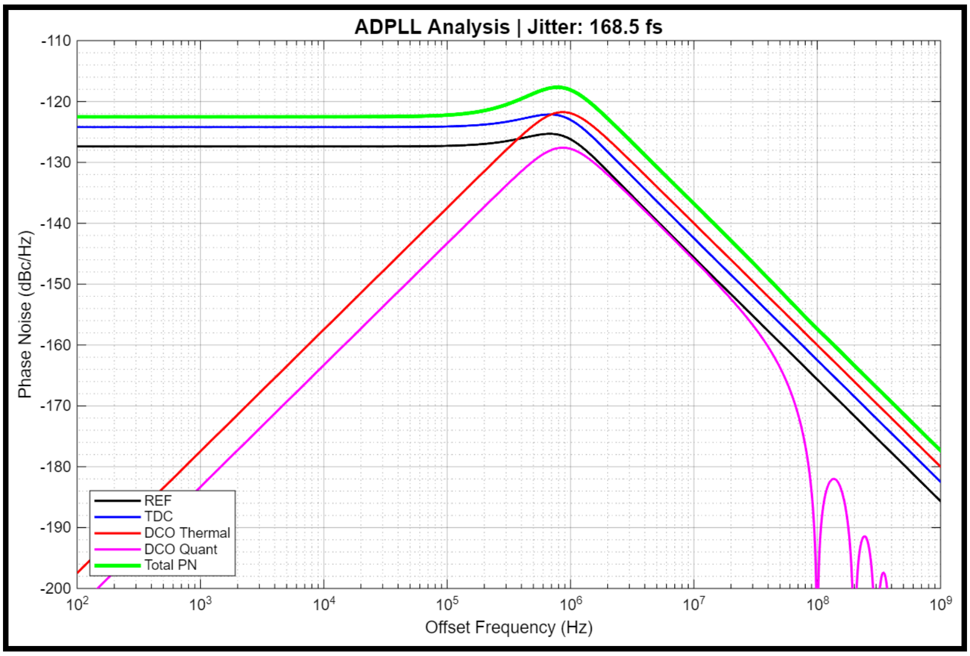

Figure 1.2: Phase noise curves.

The final values of the loop filter coefficients are:

$\rho = 2.95 * 10^{-3}$	$\alpha = 76.78 * 10^{-3}$

1.2 Answer on question C

The closed loop transfer function of the ADPLL in terms of the damping factor and the natural frequency is:

$$H(s) = N\frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{1.8}$$

The phase noise is written in the lecture slides as:

$$PM = tan^{-1}\left(2\zeta\sqrt{2\zeta^2 + \sqrt{4\zeta^4 + 1}}\right) \tag{1.9}$$

By substituting $\zeta = 1/\sqrt{2}$ we find:

$$\boxed{PM = 65^o}$$

From the closed loop transfer function, it is derived that there are two poles and one zero. That mean that the phase will not reach the -180 degrees. For this reason we cannot define a gain margin. This is an indication that the system is likely stable. Lastly, as found previously the closed loop bandwidth is:

$$\boxed{f_{BW} = 1.23MHz}$$

1.3 Answer on question D

From the theory slides I didn't observe any major differences in terms of jitter performance. A positive characteristic is that the filtering of the DCO is sharper at 40dB which is an improvement compared to the ADPLL type-I. However, it introduces also one zero in order to cancel out the extra pole of the filter. If the zero is not located close to the second pole then it will be a damping in phase (it will go lower than -90 degrees). This damping can cause some minor instabilities which can be seen as an overshoot near the bandwidth. I believe that this overshoot compensates the positive effect of the DCO noise sharper filtering. In conclusion, I believe that the effect of the second order system in terms of jitter is negligible.

In terms of stability, the indications that exist so far show a stable system. In particular, the phase margin is above 60 degrees. However, a transient analysis would give a greater insight on this issue.

1.4 Answer on question E

In this section, the magnitude and the phase of the closed-loop transfer function, which is the TDC and reference transfer function, and that of the DCO transfer function [just as an extra] in being plotted. The closed-loop transfer functions in terms of the proportional and integral gain of the loop filter are:

$$H_{CL} = N * \frac{aF_{REF} * s + \rho F_{REF}^2}{s^2 + aF_{REF} * s + \rho F_{REF}^2} \tag{1.10}$$

$$H_{DCO} = \frac{s^2}{s^2 + aF_{REF} * s + \rho F_{REF}^2} \tag{1.11}$$

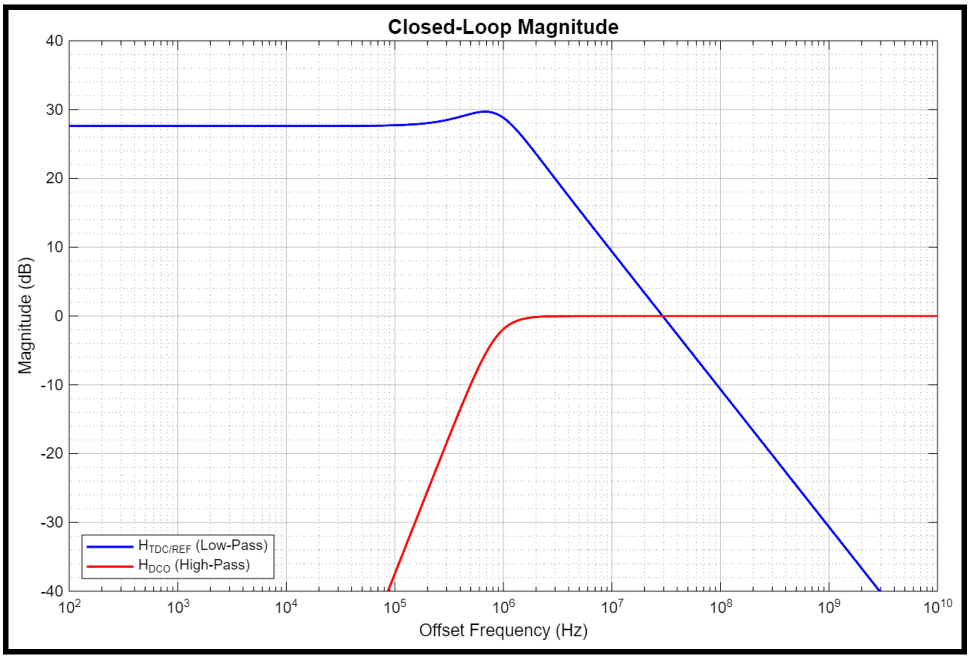

Figure 1.3: Magnitude response.

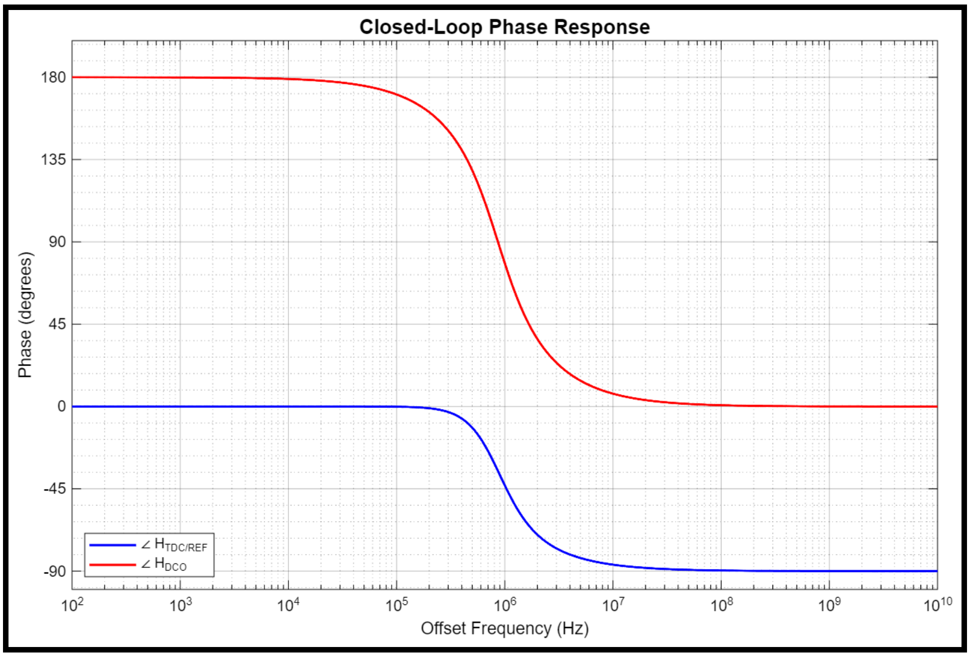

Figure 1.4: Phase response.

2 Part II [Reference and oscillator time-domain simulation]

2.1 Question A

In this question, a time-domain model for the reference oscillator is developed. To be more specific, a time model has to be built in which I will insert the noise of the reference at the signal. The purpose is to derive the phase noise from the generated time model and validate if it matches the one in the description of part 1. To build the testbench, we need to follow the steps on the theory slides. The important point is how do we insert the phase noise into the system.

From the given phase noise of the reference in dBc/Hz, the power spectral density has to be calculated at the input. The single-band phase noise in dB has to become double-band and for this reason the following expression is multiplied by 2.

$$S_{ref,in} = 2 * 10^{-155/10} = 6.32 * 10^{-16} rad^2/Hz$$

From the slides' expression, the time jitter can be found which will later be inserted in our oscillator's timestamps.

$$\sigma_{t,nf} = \frac{\sqrt{S_{ref,in} f_o}}{2\pi f_o} \tag{2.1}$$

$$\sigma_{t,nf} = 81.67fs$$

The way we insert it is by utilizing a random number with it, since the jitter is basically the standard deviation of the error in time, and hence is the statistics of the error.

At this point, we go backwards meaning that from the generated noisy signal we want to extract the standard deviation of the time error again and compute the phase error. The way it is fulfilled is by computing the standard deviation of noise error and executing this function "`[Y,f] = pwelch(phase,blackmanharris (length(phase)/20),[],[],f0,'onesided');`". This way, I find the power spectral density of the phase noise and convert in db later on. The result is:

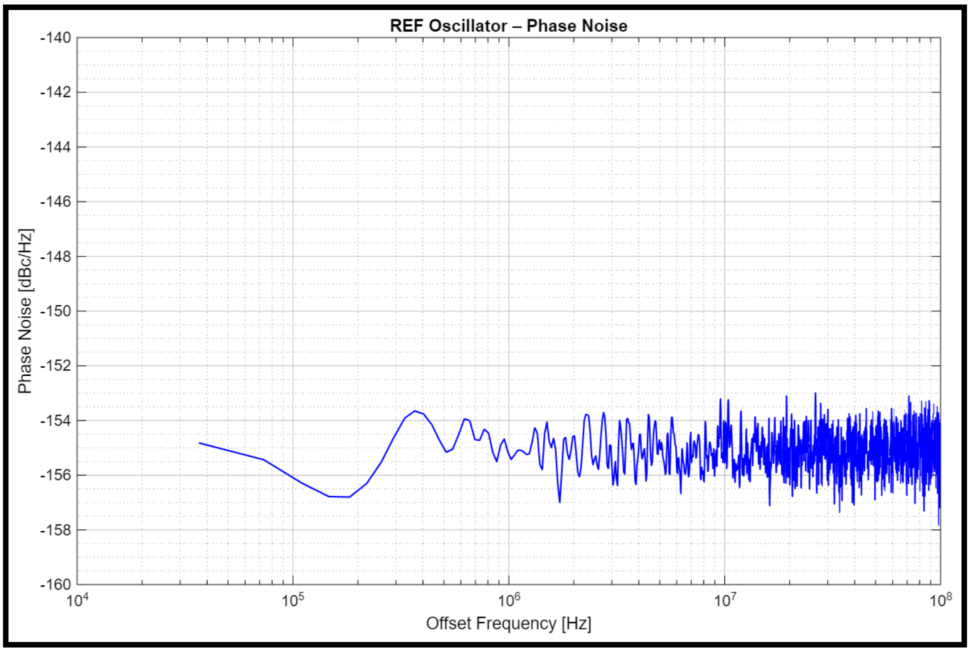

Figure 2.1: Phase noise in dB.

From figure **2.1**, it can clearly be seen that the phase error is in the vicinity of -155dBc which is what it was expected.

As an extra, I am drawing the histogram of the time error. This histogram counts the number of times a specific time error is added in the system. Since the noise is random I expect it to follow a Gaussian distribution. Moreover, I am drawing the time error in each cycle. Since there are no accumulative effects I expect the mean value of the error to center around zero.

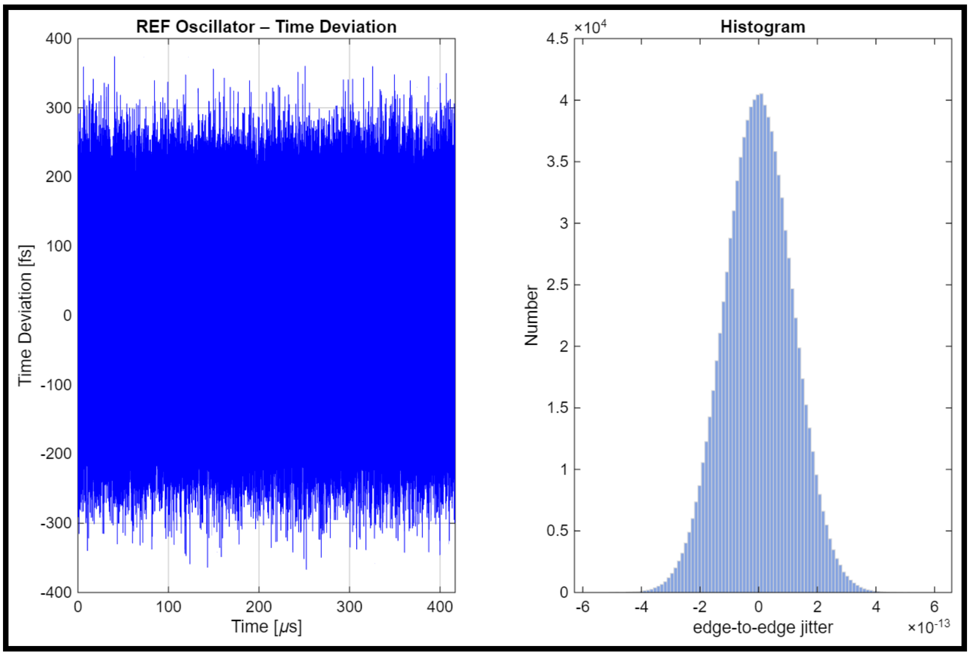

Figure 2.2: Time error over the cycles-Number of time errors added in each cycle.

2.2 Question B

In this case we consider also the thermal noise of the oscillator at an offset of 1MHz. The time jitter can be found from the theory slides expression 2.1, by substituting the power spectral density of the thermal noise at the offset frequency.

$$S_{thermal} = 2 * 10^{-120/10} = 2 * 10^{-12} rad^2/Hz$$

$$\sigma_{t,nf} = \frac{\Delta f}{f_o}\sqrt{\frac{S_{thermal}}{f_o}} \tag{2.2}$$

$$\sigma_{t,nf} = 12fs$$

In this case, the thermal noise creates an accumulative effect. The reason is that due to the thermal noise the frequency of the VCO can change slightly, it is drifting. Let's assume that it changes in the second cycle. That means that this change will be carried also in the next cycle, since the third cycle starts from where the second one ended.

The code implementation for drawing the phase noise in dB, the histogram and the time error of the cycles remain the same. In this question, the contribution of the reference and the thermal noise has taken into account and the resulted phase noise is:

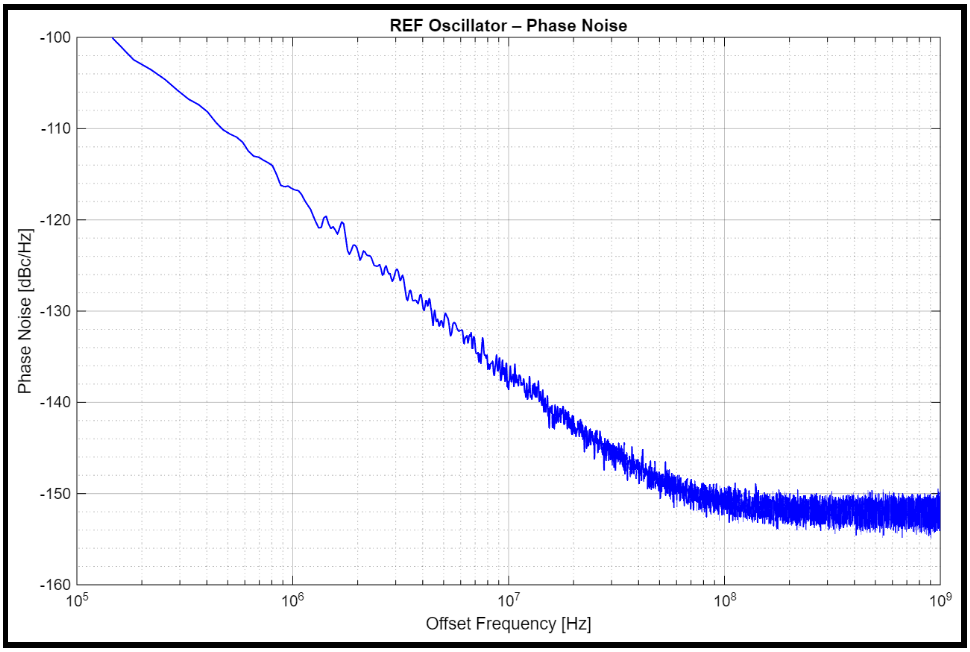

Figure 2.3: Phase noise in dB.

We observe that the phase noise has a -20dB slope at the beginning, and then it settles to -155dBc. This is what was expected since the thermal noise has a -20dB characteristic and his dominance is attenuated over frequency. It is apparent, that after 100MHz its dominance fade and the flat noise of the reference is mainly contributing.

In the next figure **2.4**, we observe that the time error is now drifting and is not centred around zero. This is what was expected, since the thermal noise of the VCO adds the accumulative effect. The histogram show again a Gaussian shaped curve, since this is referring to the time error added in each cycle excluding the accumulative error.

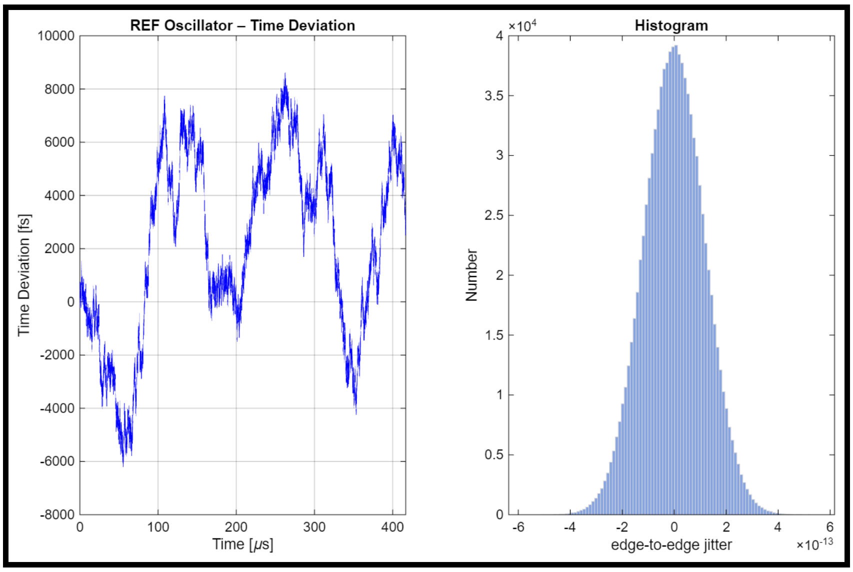

Figure 2.4: Time error over the cycles-Number of time errors added in each cycle.

3 Part III [ADPLL time-domain simulation]

3.1 Question A

At the beginning, a set of samples was set for the locking process of the ADPLL and a bigger set of samples was set for the condition where the ADPLL is locked. The vector which contained the information in order to perform the simulations in the next questions was "CKV_locked".

At first, the reference signal was created:

```
for n = 1:N_total
    t_ref(n) = n * T_ref + sigma_t_ref * randn(1);
end
```

Afterwards, the early condition where no cycles have arrived yet was initialized.

```
% 8. Initializing the output of the divider
for n = 1:N
    edge_ptr = edge_ptr + 1;
    t_ckv = t_ckv + T_o + sigma_t_dco * randn(1) - K_T*
       OTW_init;
end
t_div(1) = t_ckv; % First divider output after 24 ref cycles.
```

In the for loop, in which the time model is running and save the data at the vector "CKV_locked", the time error and the output code of the TDC was found. Moreover, the frequency in which the DCO is running when a new cycle is arriving from the reference was initialized. This frequency is used for the calculation of the input value of the filter.

For the loop filter, the expressions inside the block, regarding the implemention of the proportional and the integral gain, were followed. In the next step, the input word code for the DCO was generated and the instantaneous values of the DCO frequency and period was found, in order to plot the relevant waveforms. At the end of the main for loop the process of the data collection was implemented.

```
 if k < N_total
        for n = 1:N
            edge_ptr = edge_ptr + 1; % Count and indexing of
               the DCO timestamps
            t_ckv    = t_ckv + T_o + sigma_t_dco * randn(1) -
                K_T * OTW(k);

            % Store CKV timestamps only for the locked
               section
            if k >= N_locking
                locked_idx = (k - N_locking)*N + n;
                if locked_idx >= 1 && locked_idx <= numel(
                   CKV_locked)
                    CKV_locked(locked_idx) = t_ckv;
                end
```

```
            end
        end
        t_div(k+1) = t_ckv; % 24th DCO timestamp becomes the
           divider timestamp
end
```

The if statement at the beginning checks if the time model has reached is maximum capacity. Then it starts with the genetation of the next DCO timestamps. Afterwards, if the time needed for the locking is passed it starts saving the timestamps that the DCO produces in the "`CKV_locked`" vector.

3.2 Question B

3.2.1 Plot 1

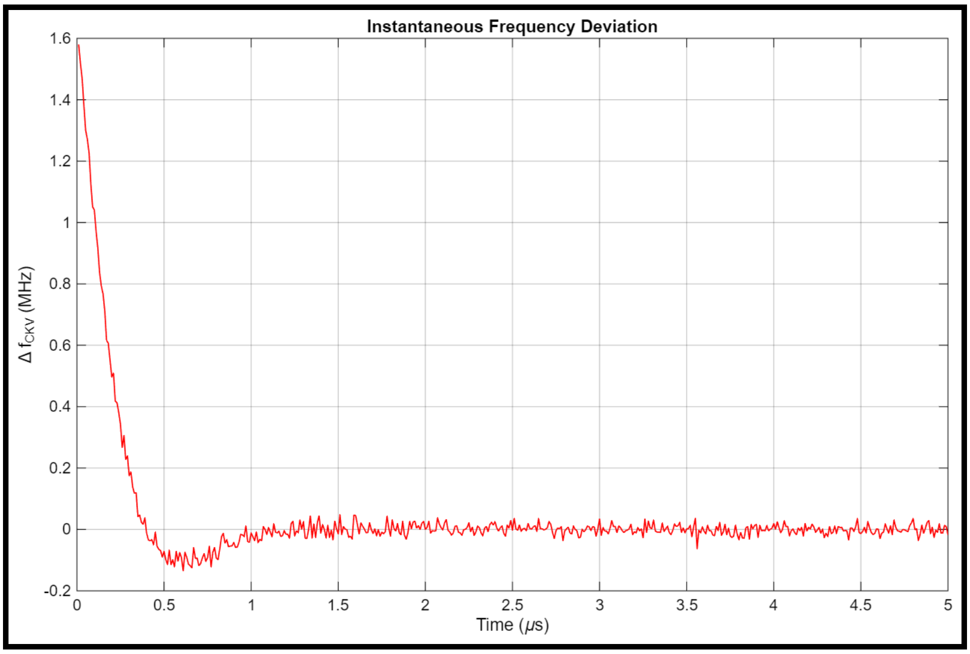

Figure 3.1: Instantaneous frequency deviation.

The settling is adequatly fast, around 2us. This comes from the choice of ζ which was decided to be $1/\sqrt{2}$. Even though it might affect the stability due to the complex poles, we can clearly see that there is only an overshoot and no damping. Selecting the coefficient ζ>1, we could have better stability, even speed if ζ was close to one, but the bigger the ζ is the closer the one of the two poles of the second order system goes closer to the origin. This would make the ADPLL a lot slower that the positive effect of stability isn't that important any more. Lastly, in the following graphs we observe also minimum damping and fast settling.

3.2.2 Plot 2

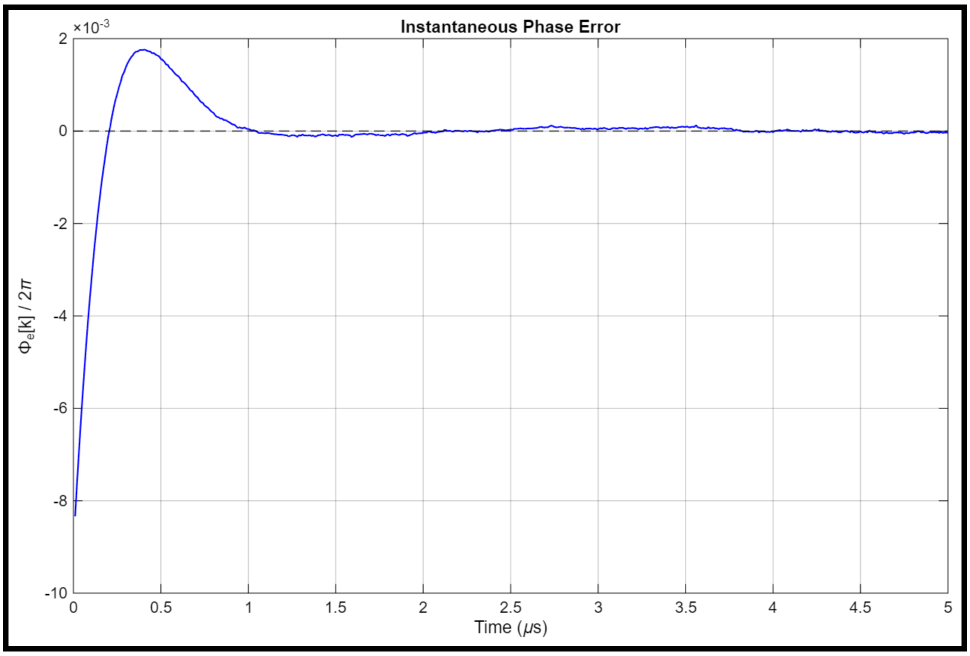

Figure 3.2: Instantaneous phase error.

It does not follow the theoritical waveform of the slides.

3.2.3 Plot 3

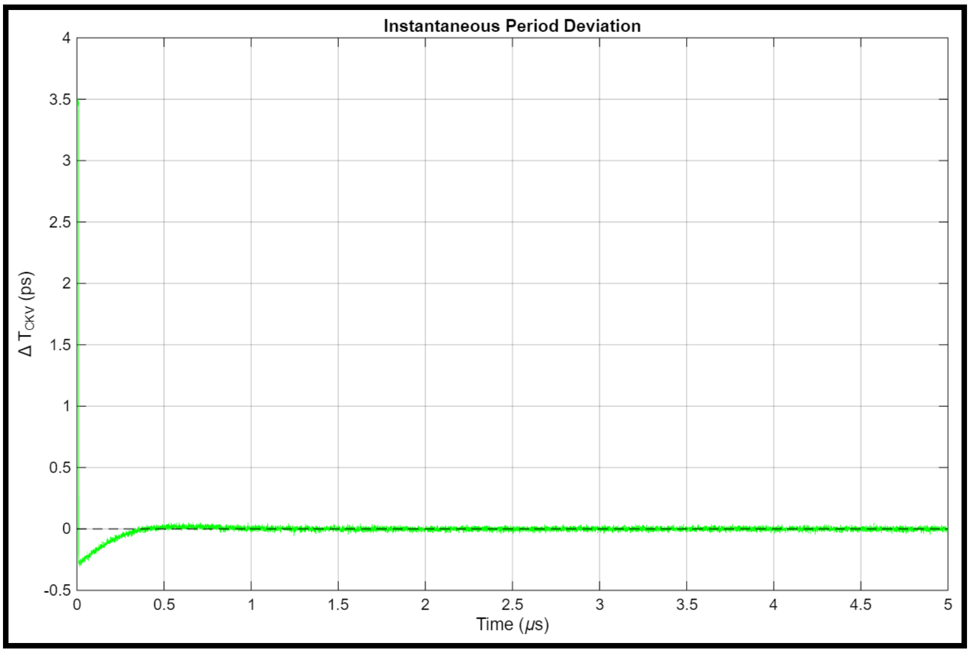

Figure 3.3: Instantaneous period deviation.

3.3 Question C-D

In figure **3.4**, the phase noise is presented.

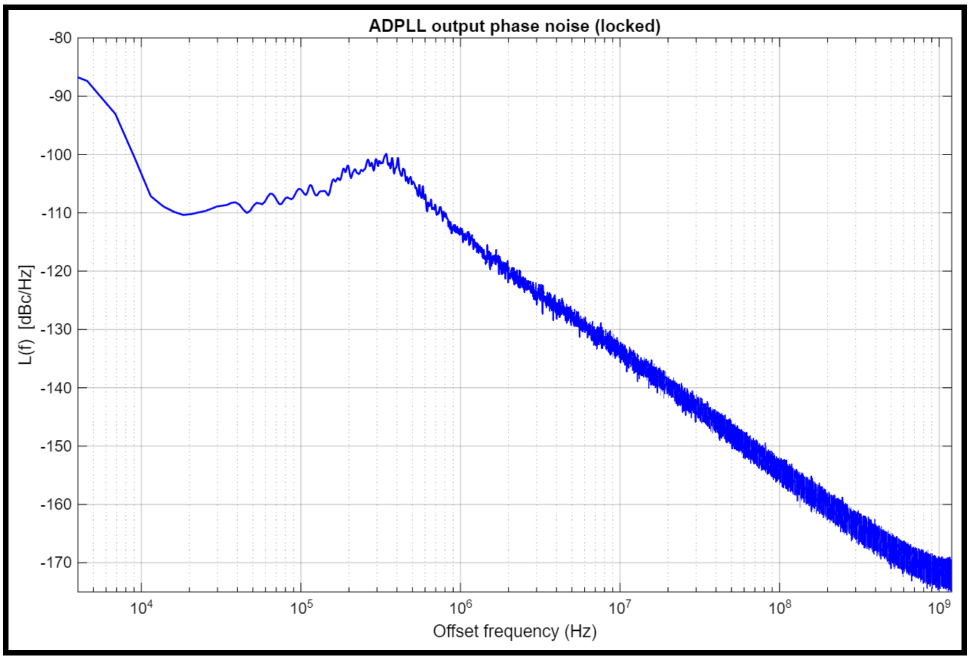

Figure 3.4: Phase noise.

Even though the phase noise derived by the time-domain model somewhat follows the expected total phase noise presented in figure **1.2**, it is nowhere close to follow the right behaviour in band. The waveform is way above the -120dBc, where I was expected the phase noise to land inband according to the figure **1.2**. As a result, the developed time domain model is inadequate to simulate an ADPLL, and more effort is needed to build the code of part 3, correctly. This becomes more apparent by the calculation of the jitter noise:

$$\boxed{\sigma_\tau = 0.516ps}$$